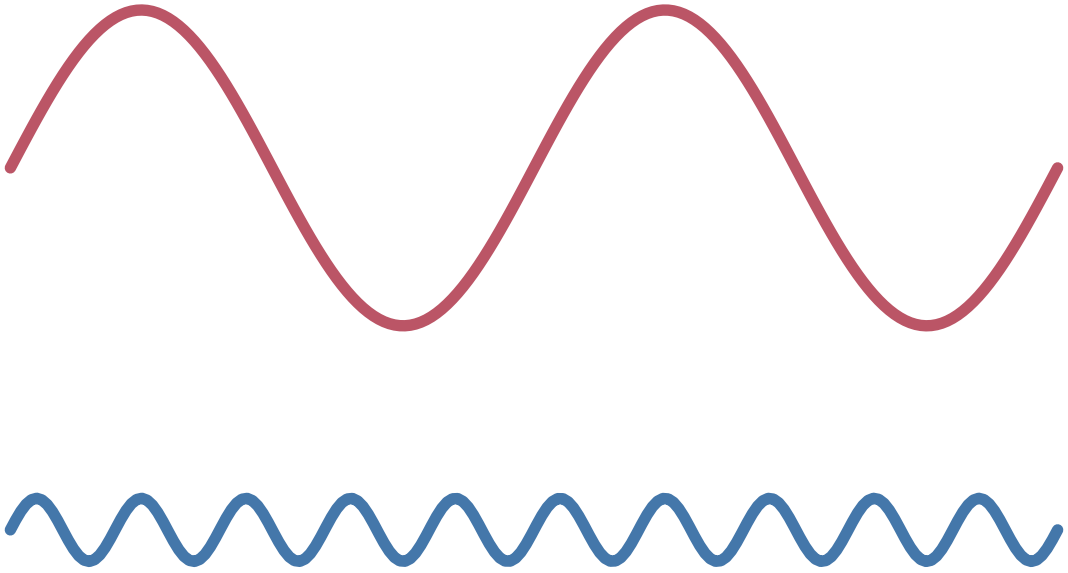
# Harmonics & Resonance

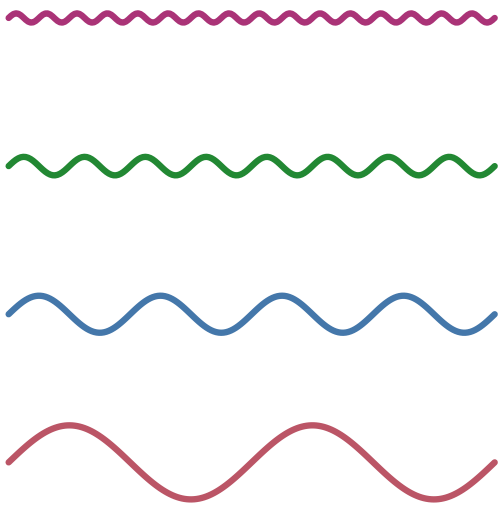
Dr. Josef Fruehwald

## Complex Sounds



Complex Sounds

Complex Sounds

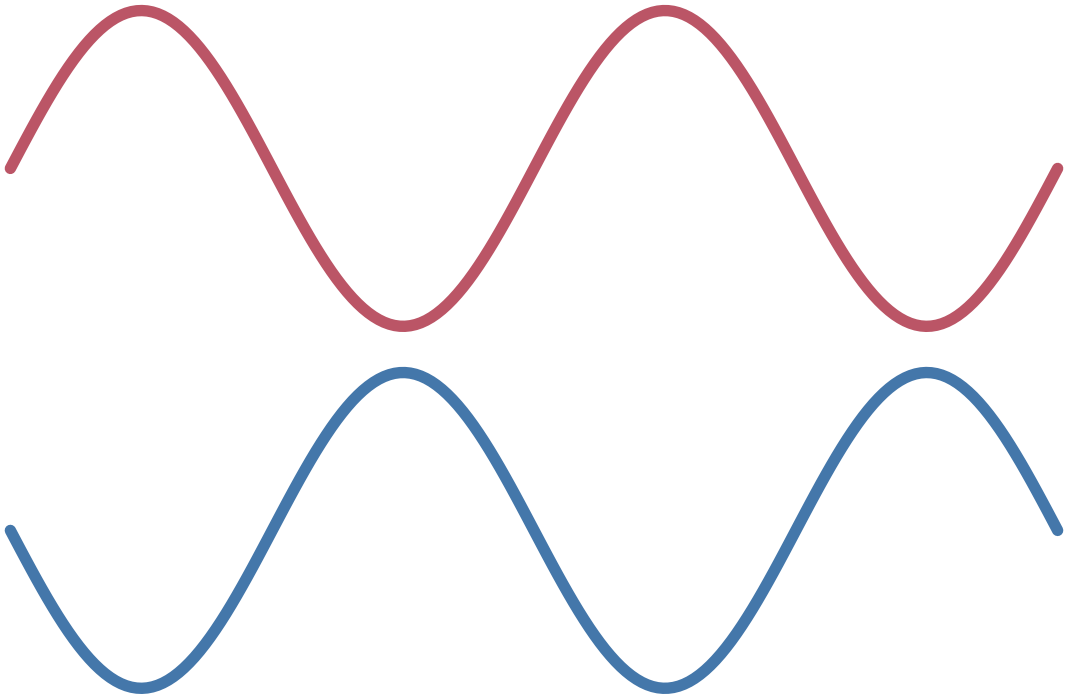




Complex sounds

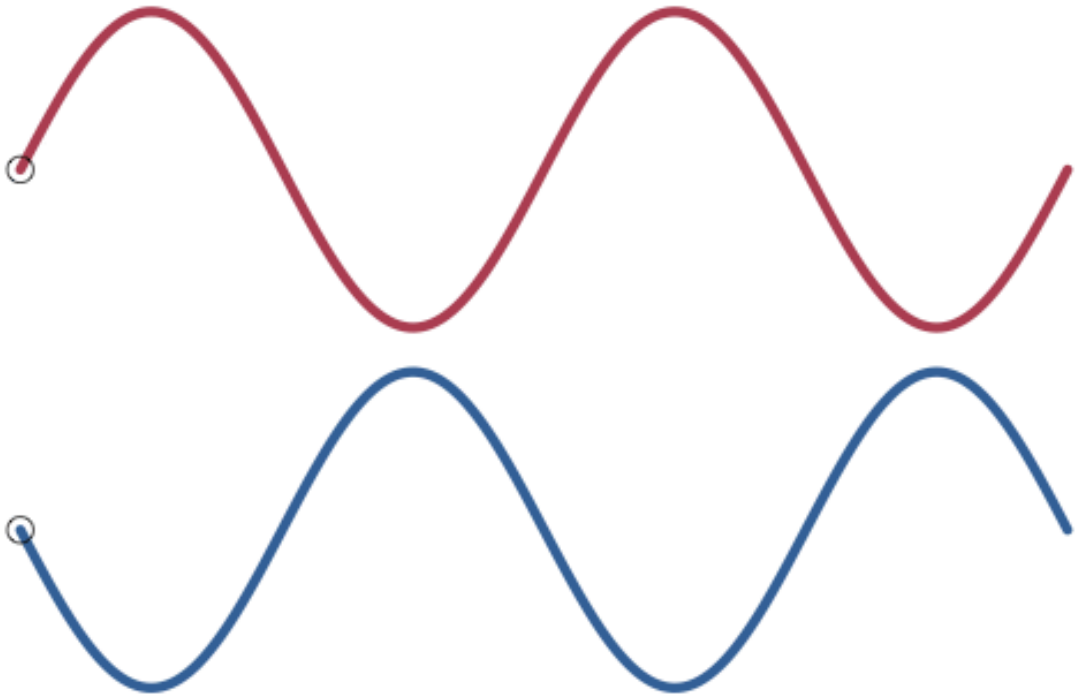
Complex Sounds

Interference

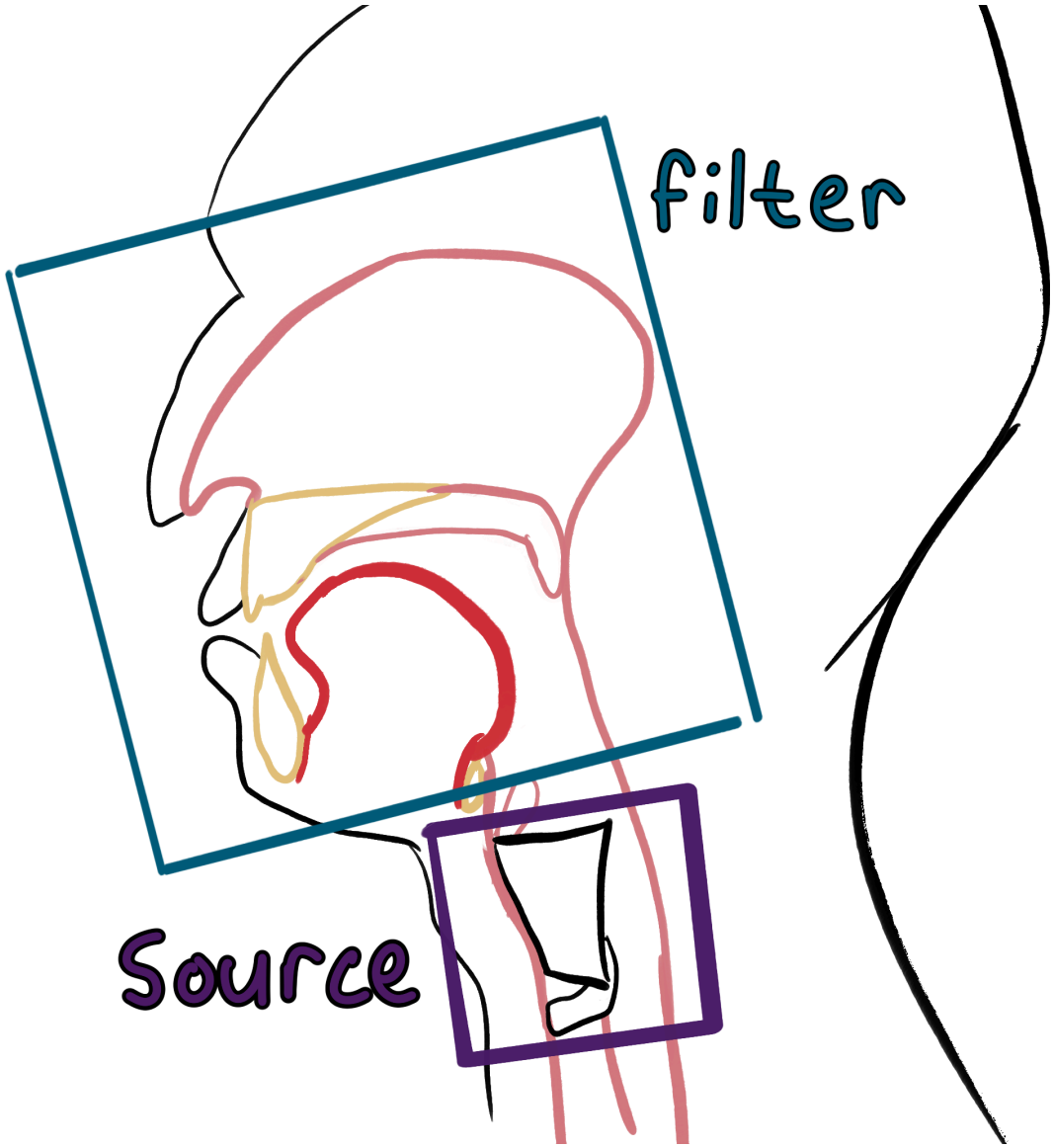


# Complex Sounds

## Interference

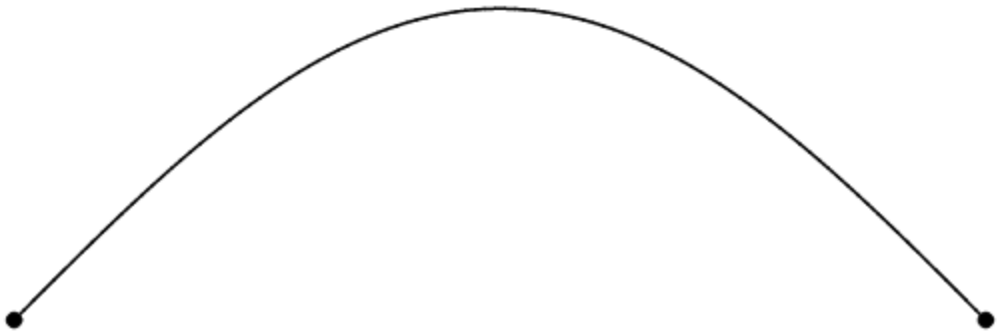


## The Source-Filter Model




## Fundamental Frequency - From the Source

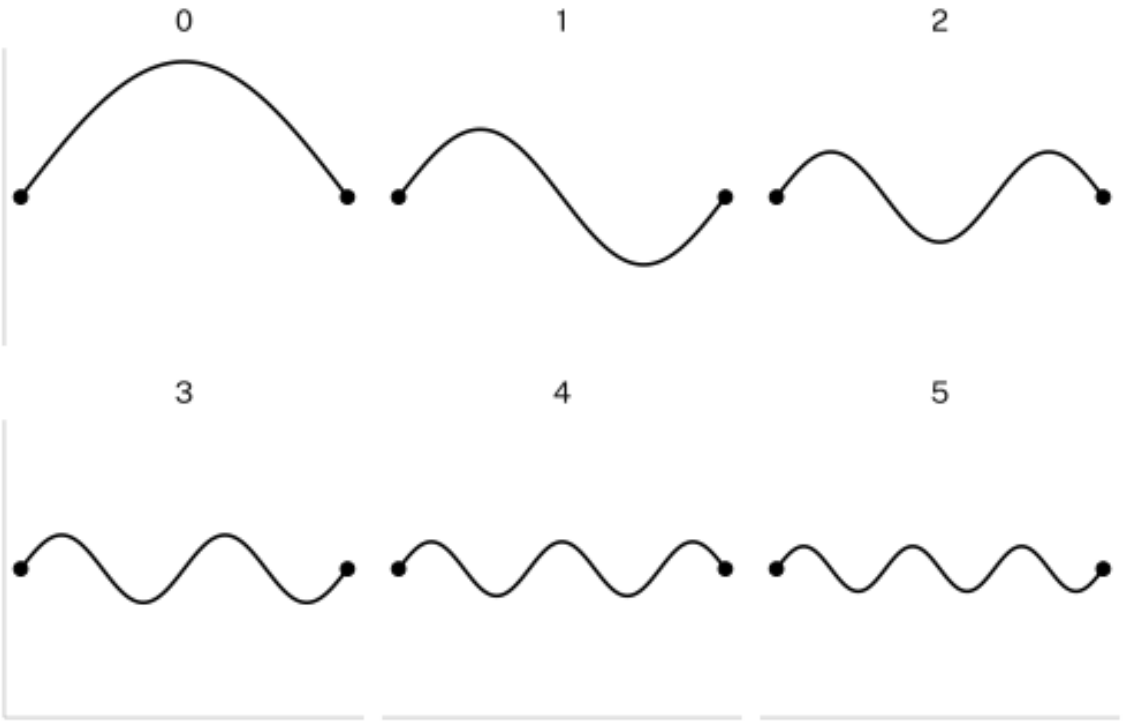
The wavelength is proportional to the string length.



## Fundamental Frequency - From the Source

Wave Length = 2Length

## Harmonics - From the Source




## Harmonics - From the Source

$$WL_n = \frac{2L}{n}$$

- When $n$ = 1, $WL$ is the fundamental frequency.
- For integers > 1, $WL$ is a harmonic.
- Every harmonic wavelength is the fundamental divided by a whole number integer.
- Every harmonic *frequency* is a whole number *multiple* of the fundamental frequency.

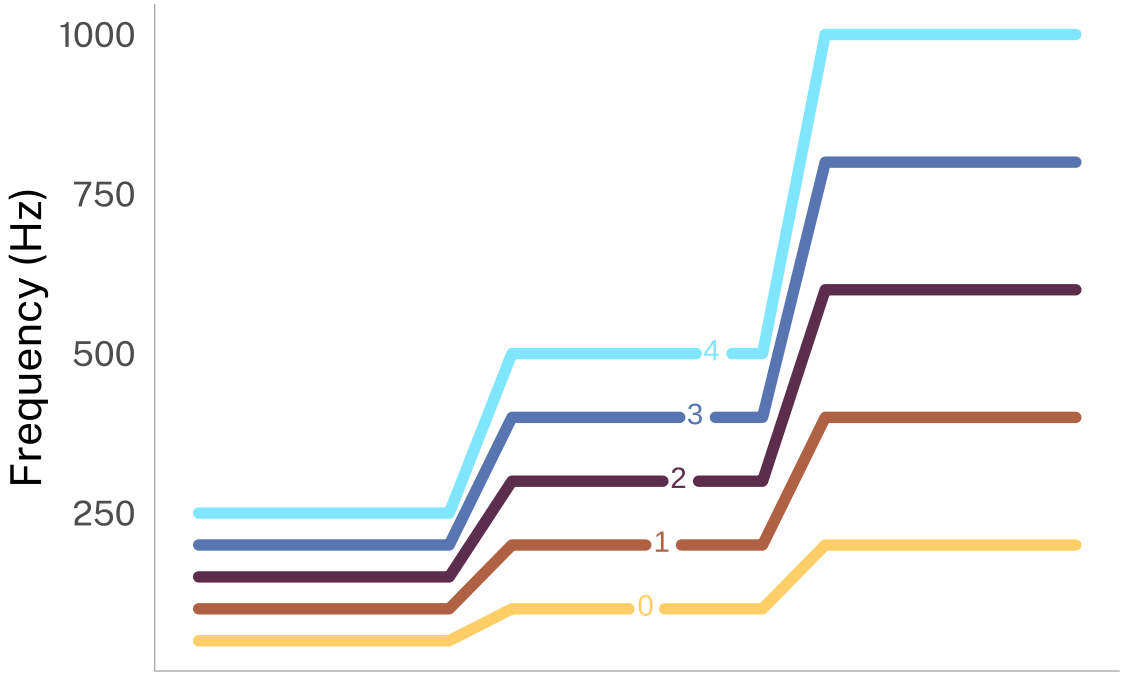
# Harmonics

## Wavelength to Frequency

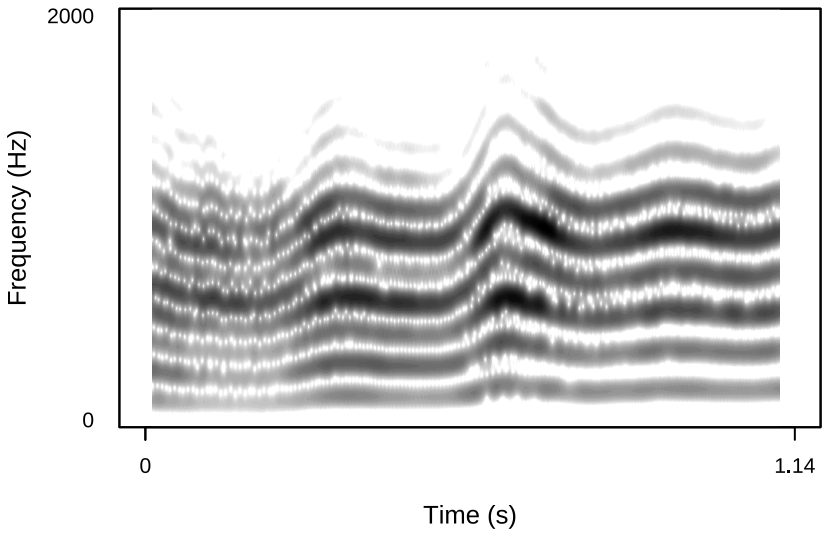
| Harmonic | Wave-length | Frequency |
|---|---|---|
| 0 | 686 cm | 50 Hz |
| 1 | 1372 cm | 100 Hz |
| 2 | 2058 cm | 150 Hz |
| 3 | 2744 cm | 200 Hz |
| 4 | 3430 cm | 250 Hz |

| Harmonic | Wave-length | Frequency |
|---|---|---|
| 0 | 343 cm | 100 Hz |
| 1 | 686 cm | 200 Hz |
| 2 | 1029 cm | 300 Hz |
| 3 | 1372 cm | 400 Hz |
| 4 | 1715 cm | 500 Hz |

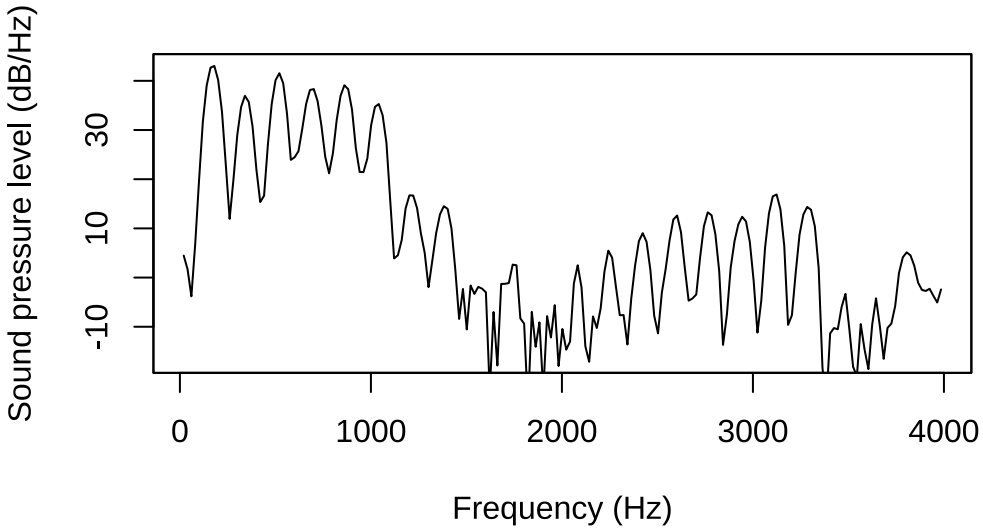
## Harmonics



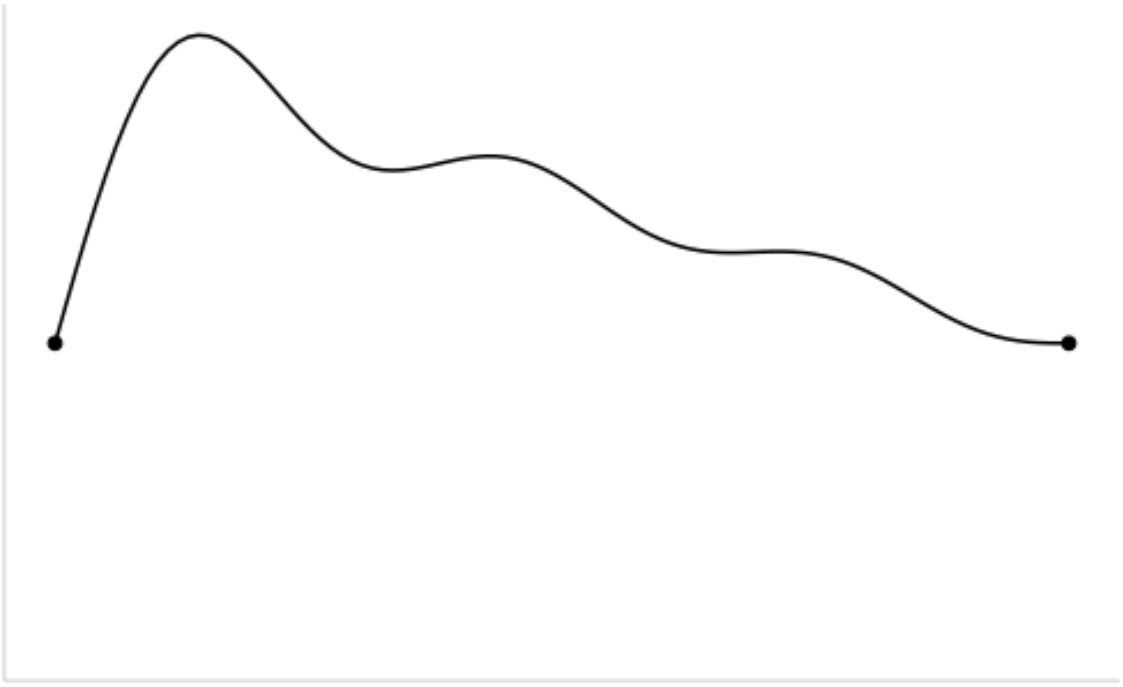
## Harmonics



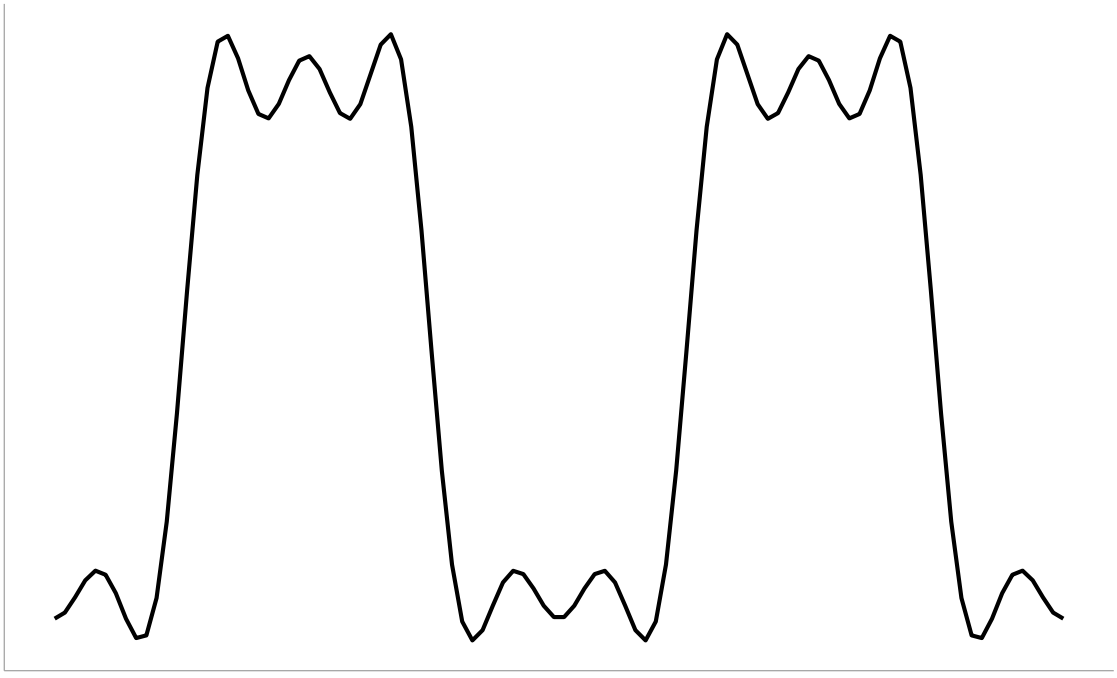

## Harmonics



Harmonics

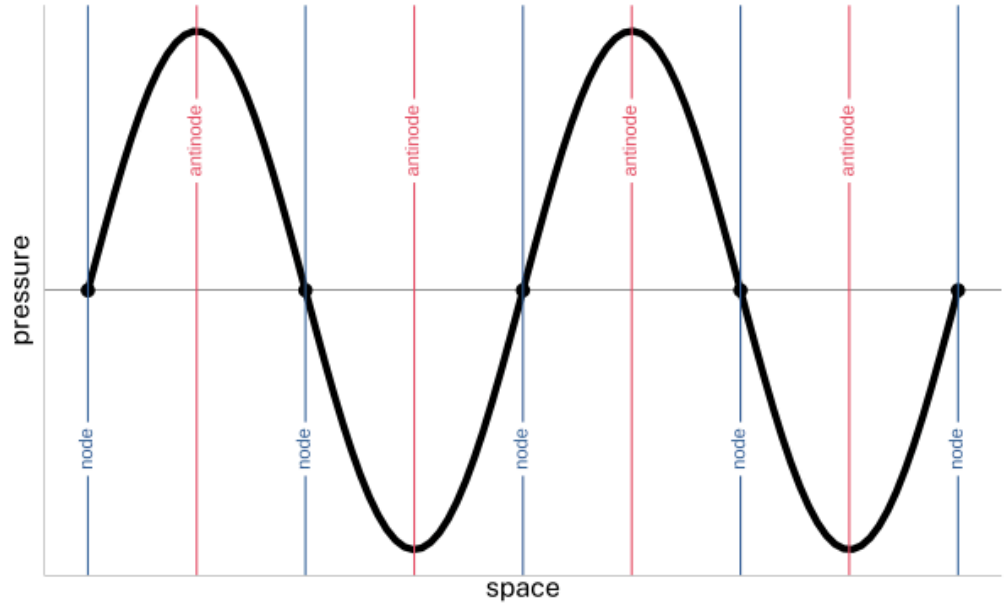


Harmonics



# Standing Waves

## Nodes and Antinodes

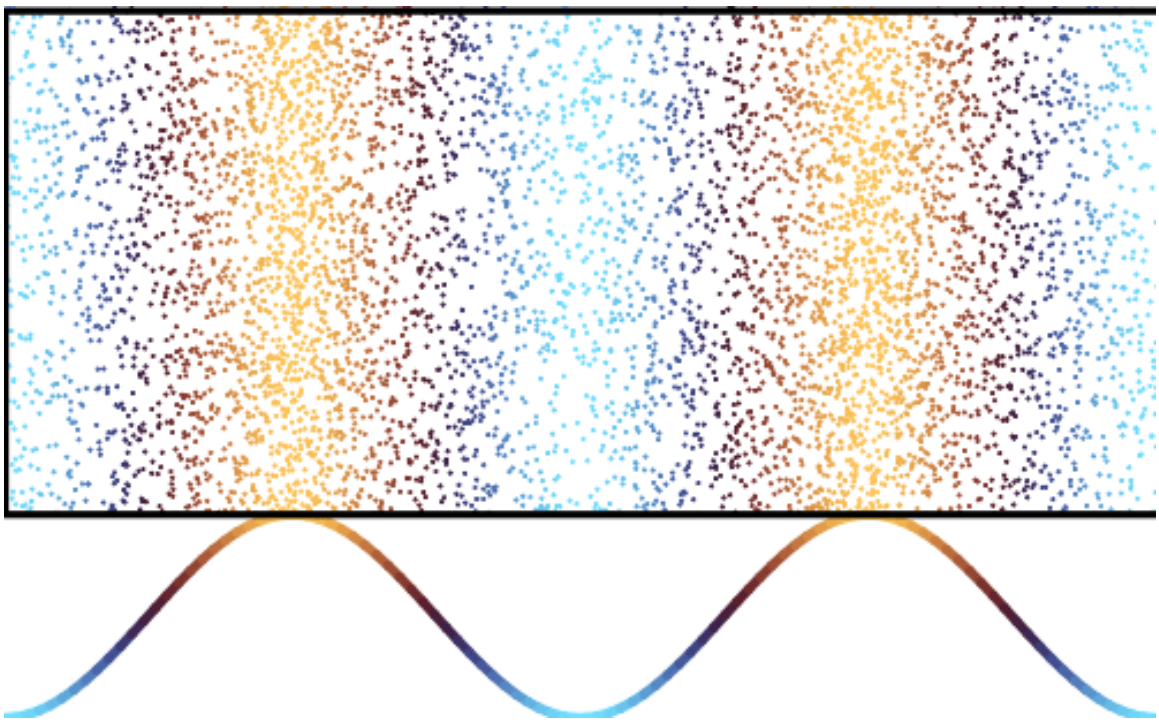



## From the source to the filter

- The physical activity of the vocal folds generates the fundamental frequency and the harmonics.
- Then, the air pressure waves propagate through your vocal tract, which introduces *resonances*.

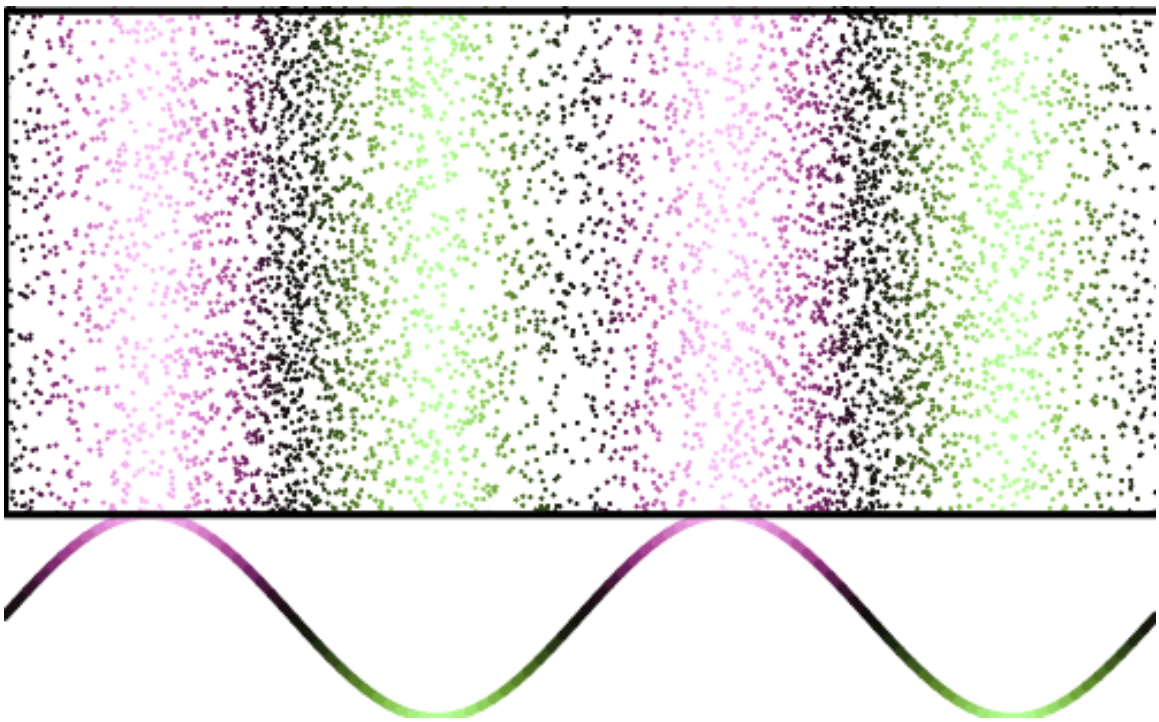
# Resonance - From the Filter

## Pressure



## Pressure & Displacement

### Displacement

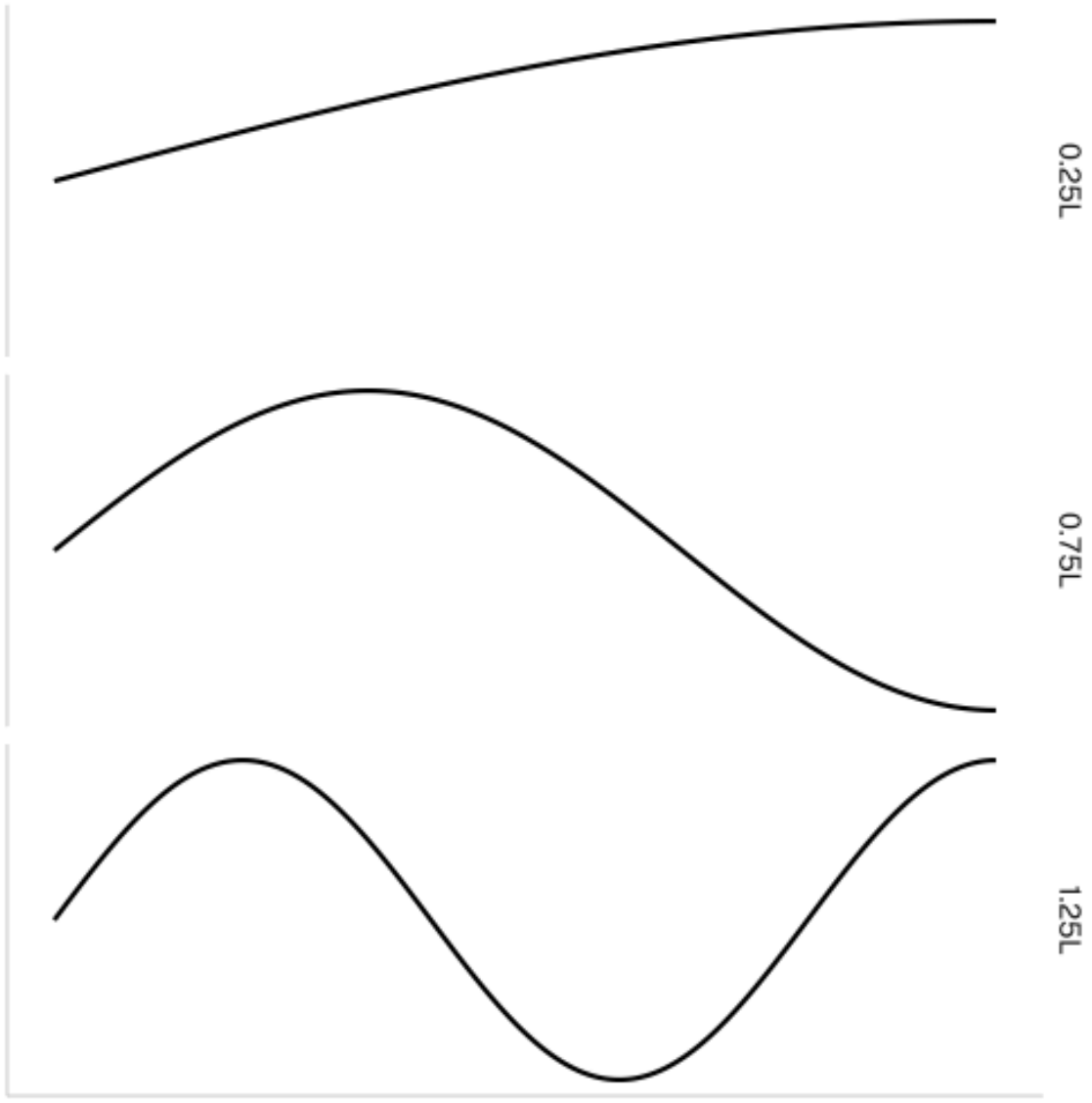


## Resonance - From the Filter

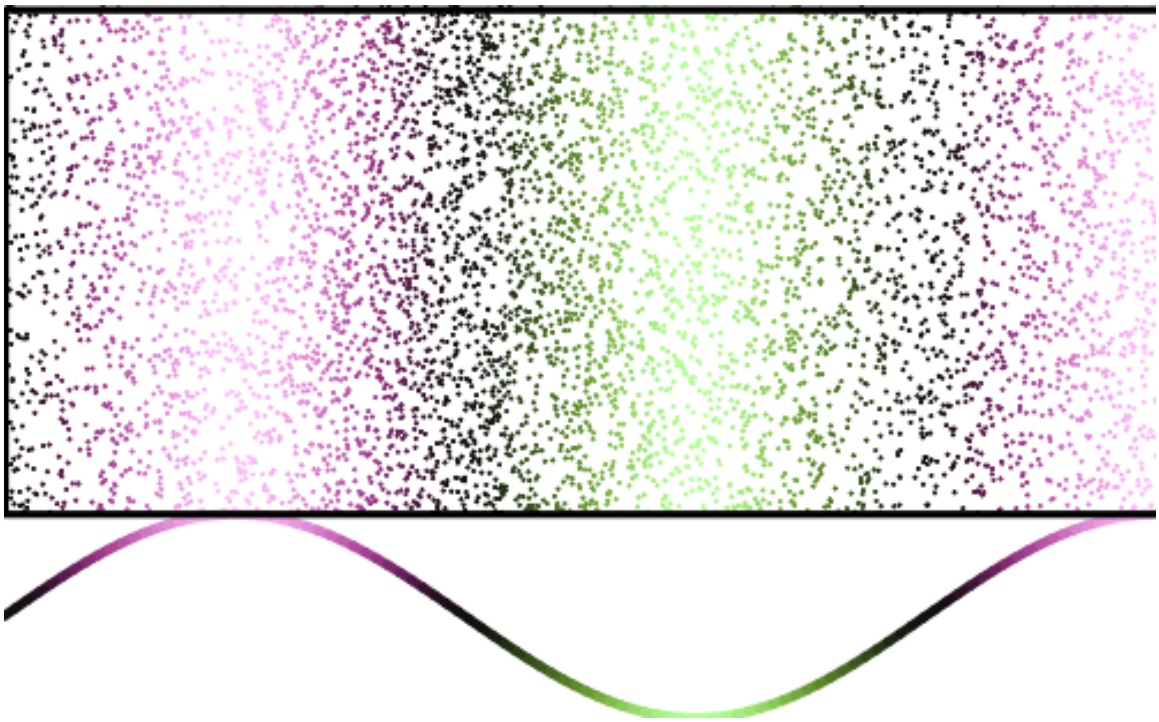
https://www.youtube.com/watch?v=dihQuwrf9yQ

## Resonance - From the filter

Wavelengths that put a displacement antinode at the lips are resonant frequencies.


0.25L
0.75L
1.25L

## Resonance - From the Filter



## Resonance - From the Filter

$$F_n = \frac{(2n-1)c}{4L}$$

$$L = \frac{(2n-1)c}{4F_n}$$

- $n$ = resonant frequency
- $c$ = speed of sound